

CBSE Class 12 Chemistry — Electrochemistry

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark Questions (MCQs / Assertion-Reason / Very Short Answer)

- Q1.** Which of the following solutions will have the highest conductivity at 298 K? (a) 0.01 M HCl (b) 0.1 M HCl (c) $0.01\text{ M CH}_3\text{COOH}$ (d) $0.1\text{ M CH}_3\text{COOH}$ [PYQ 2023] [Confidence: Very High]
- Q2.** Which of the following solutions of KCl will have the highest value of molar conductivity? (a) 0.01 M (b) 1 M (c) 0.5 M (d) 0.1 M [PYQ 2023] [Confidence: Very High]
- Q3.** How much charge in terms of Faraday is required for the reduction of 1 mol of Zn^{2+} to Zn ? (a) 1 F (b) 2 F (c) 3 F (d) 0.5 F [PYQ 2015] [Confidence: Very High]
- Q4.** How much charge in terms of Faraday is required to reduce one mole of MnO_4^- to Mn^{2+} ? (a) 1 F (b) 3 F (c) 5 F (d) 7 F [PYQ 2020] [Confidence: Very High]
- Q5.** How many coulombs of electricity are required for the oxidation of 1 mole of H_2O to O_2 ? (a) 96500 C (b) 193000 C (c) 289500 C (d) 48250 C [PYQ 2020] [Confidence: High]
- Q6.** The limiting molar conductivity Λ_m° for CH_3COOH can be calculated if the values of the following are given: (a) $\Lambda_m^\circ \text{HCl}$, $\Lambda_m^\circ \text{KCl}$ and $\Lambda_m^\circ \text{CH}_3\text{COOK}$ (b) $\Lambda_m^\circ \text{NaCl}$, $\Lambda_m^\circ \text{KCl}$ and $\Lambda_m^\circ \text{CH}_3\text{COONa}$ (c) $\Lambda_m^\circ \text{H}_2\text{SO}_4$, $\Lambda_m^\circ \text{Na}_2\text{SO}_4$ and $\Lambda_m^\circ \text{CH}_3\text{COONa}$ (d) Both (a) and (c) [PYQ 2025-26 SQP] [Confidence: Very High]
- Q7.** The relationship between conductivity (κ), cell constant (G^*), and observed resistance (R) of a conductivity cell is given by: (a) $\kappa = G^*/R$ (b) $\kappa = G^* \times R$ (c) $\kappa = R/G^*$ (d) $\kappa = G^* + R$ [Practice] [Confidence: High]
- Q8.** What is the SI unit of molar conductivity (Λ_m)? (a) S m^{-1} (b) $\text{S cm}^2 \text{mol}^{-1}$ (c) Ωcm^{-1} (d) S mol^{-1} [Practice] [Confidence: Very High]
- Q9.** E_{cell}° for a redox reaction is 2.71 V . What will be the direction of flow of current when an external opposite potential of 3.0 V is applied? (a) Cathode to Anode (b) Anode to Cathode (c) No current flows (d) Current oscillates [PYQ 2019] [Confidence: Very High]
- Q10.** Why does the cell voltage of a mercury cell remain constant during its lifetime? [PYQ 2022] [Confidence: Very High]
- Q11.** On the basis of E° values, O_2 gas should be liberated at the anode, but Cl_2 gas is liberated at the anode during the electrolysis of aqueous NaCl . Give a reason. [PYQ 2020] [Confidence: High]
- Q12.** Name the type of cell which was used in the Apollo space programme. [PYQ 2020] [Confidence: High]
- Q13.** Write the cell representation for the Daniell cell. [Practice] [Confidence: Medium]
- Q14.** What is the value of Faraday's constant and what does it represent? [Practice] [Confidence: Medium]
- Q15.** What is the unit of the cell constant? [Practice] [Confidence: Medium]
- Q16.** Assertion (A): Conductivity of an electrolyte decreases with a decrease in concentration. Reason (R): Number of ions per unit volume that carry the current in a solution decreases on dilution. (a) Both (A) and (R) are true and (R) is the correct explanation of (A). (b) Both (A) and (R) are true but (R) is not the correct explanation of (A). (c) (A) is true but (R) is false. (d) (A) is false but (R) is true. [Practice] [Confidence: Very High]
- Q17.** Assertion (A): The molar conductivity of a weak electrolyte increases steeply on dilution. Reason (R): The degree of dissociation of a weak electrolyte increases significantly with dilution. (a) Both (A) and (R) are true and (R) is the correct explanation of (A). (b) Both (A) and (R) are true but (R) is not the correct explanation of (A). (c) (A) is true but (R) is false. (d) (A) is false but (R) is true. [Practice] [Confidence: Very High]
- Q18.** Assertion (A): At equilibrium, the standard cell potential (E_{cell}°) of a galvanic cell becomes zero. Reason (R): At equilibrium, the reaction quotient (Q_c) becomes equal to the equilibrium constant (K_c). (a) Both (A) and (R) are true and (R) is the correct explanation of (A). (b) Both (A) and (R) are true but (R) is not the correct explanation of (A). (c) (A) is true but (R) is false. (d) (A) is false but (R) is true. [Practice] [Confidence: High]
- Q19.** Is it possible to identify whether a given solution is 1 M NaOH or 1 M HCl using only a simple conductivity meter with LEDs? Justify your answer. [PYQ 2025-26 SQP] [Confidence: High]
- Q20.** Out of 0.1 M HCl and $0.1\text{ M CH}_3\text{COOH}$, which solution will have higher conductivity and why? [Practice] [Confidence: High]

2-Mark Questions (Short Answer)

- Q21.** State Faraday's first law of electrolysis. Write its mathematical expression. [PYQ 2015] [Confidence: Very High]
- Q22.** Give two points of difference between an electrolytic cell and an electrochemical (galvanic) cell. [PYQ 2020] [Confidence: Very High]
- Q23.** Explain why the conductivity (κ) of a solution decreases with dilution, whereas its molar conductivity (Λ_m) increases with dilution. [PYQ 2015] [Confidence: Very High]
- Q24.** Define the term degree of dissociation (α). Write an expression that relates the molar conductivity of a weak electrolyte to its degree of dissociation. [PYQ 2015] [Confidence: High]
- Q25.** Write the name of the cell which is generally used in hearing aids. Write the reactions taking place at the anode and the cathode of this cell. [PYQ 2017] [Confidence: High]
- Q26.** Define a fuel cell. Write any two advantages of fuel cells over ordinary batteries. [PYQ 2018] [Confidence: Very High]
- Q27.** The standard reduction potentials are $E_{Zn^{2+}/Zn}^\circ = -0.76\text{ V}$ and $E_{Cu^{2+}/Cu}^\circ = +0.34\text{ V}$. Calculate the standard Gibbs free energy change ($\Delta_r G^\circ$) for the reaction: $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$. ($1\text{ F} = 96500\text{ C mol}^{-1}$) [PYQ 2014] [Confidence: Very High]
- Q28.** State Kohlrausch's law of independent migration of ions. How is it useful in determining the limiting molar conductivity of a weak electrolyte? [PYQ 2014] [Confidence: Very High]
- Q29.** Calculate the limiting molar conductivity (Λ_m°) for $MgCl_2$ from the data given below:
 $\lambda_m^\circ Mg^{2+} = 106.0\text{ S cm}^2\text{ mol}^{-1}$ and $\lambda_m^\circ Cl^- = 76.3\text{ S cm}^2\text{ mol}^{-1}$. [CBSE SQP 2023] [Confidence: High]
- Q30.** Estimate the minimum potential difference needed to reduce Al_2O_3 at 500°C . The free energy change for the decomposition reaction $\frac{2}{3}Al_2O_3 \rightarrow \frac{4}{3}Al + O_2$ is $\Delta G = +960\text{ kJ}$. ($1\text{ F} = 96500\text{ C mol}^{-1}$) [PYQ 2014] [Confidence: Medium]
- Q31.** In the plot of molar conductivity (Λ_m) vs square root of concentration ($\sqrt{c}$), two curves are obtained for electrolytes A and B. Curve A is a straight line, while Curve B shows a steep increase at low concentrations. Predict the nature of electrolytes A and B. What happens on extrapolation of Λ_m to concentration approaching zero for both? [PYQ 2019] [Confidence: High]
- Q32.** Can we construct an electrochemical cell with two half-cells both composed of $ZnSO_4$ solution and zinc electrodes? Explain your answer. [CBSE SQP 2023] [Confidence: Medium]

3-Mark Questions (Short Answer)

- Q33.** Write the Nernst equation and calculate the e.m.f. of the following cell at 298 K:
 $Zn(s)|Zn^{2+}(0.1\text{ M})||Ag^+(0.01\text{ M})|Ag(s)$ Given: $E_{Zn^{2+}/Zn}^\circ = -0.76\text{ V}$, $E_{Ag^+/Ag}^\circ = +0.80\text{ V}$. [PYQ 2020] [Confidence: Very High]
- Q34.** Calculate the maximum work (ΔG°) and $\log K_c$ for the given reaction at 298 K:
 $Ni(s) + 2Ag^+(aq) \rightleftharpoons Ni^{2+}(aq) + 2Ag(s)$ Given: $E_{Ni^{2+}/Ni}^\circ = -0.25\text{ V}$, $E_{Ag^+/Ag}^\circ = +0.80\text{ V}$, $1\text{ F} = 96500\text{ C mol}^{-1}$. [PYQ 2020] [Confidence: Very High]
- Q35.** The resistance of a conductivity cell filled with $0.1\text{ mol L}^{-1}\text{ KCl}$ solution is $100\ \Omega$. If the resistance of the same cell when filled with $0.02\text{ mol L}^{-1}\text{ KCl}$ solution is $520\ \Omega$, calculate the conductivity and molar conductivity of $0.02\text{ mol L}^{-1}\text{ KCl}$ solution. (Given: Conductivity of $0.1\text{ mol L}^{-1}\text{ KCl}$ solution is $1.29 \times 10^{-2}\ \Omega^{-1}\text{ cm}^{-1}$). [PYQ 2014] [Confidence: Very High]
- Q36.** The limiting molar conductivities (Λ_m°) for $NaCl$, HCl , and CH_3COONa are 126.4 , 425.9 , and $91.0\text{ S cm}^2\text{ mol}^{-1}$ respectively. Calculate the limiting molar conductivity of acetic acid (CH_3COOH). [PYQ 2022] [Confidence: Very High]
- Q37.** Calculate the degree of dissociation (α) of 0.0024 M acetic acid if the conductivity of this solution is $8.0 \times 10^{-5}\text{ S cm}^{-1}$. Given: $\lambda_{H^+}^\circ = 349.6\text{ S cm}^2\text{ mol}^{-1}$ and $\lambda_{CH_3COO^-}^\circ = 40.9\text{ S cm}^2\text{ mol}^{-1}$. [PYQ 2019] [Confidence: Very High]
- Q38.** Calculate the mass of Ag deposited at the cathode when a current of 2 amperes was passed through a solution of $AgNO_3$ for 15 minutes . (Given: Molar mass of $Ag = 108\text{ g mol}^{-1}$, $1\text{ F} = 96500\text{ C mol}^{-1}$) [PYQ 2017] [Confidence: Very High]

Q39. A steady current of **2 amperes** was passed through two electrolytic cells X and Y connected in series containing electrolytes FeSO_4 and ZnSO_4 until **2.8 g** of Fe deposited at the cathode of cell X. How long did the current flow? Calculate the mass of Zn deposited at the cathode of cell Y. (Molar mass: $\text{Fe} = 56 \text{ g mol}^{-1}$, $\text{Zn} = 65.3 \text{ g mol}^{-1}$, $1 \text{ F} = 96500 \text{ C mol}^{-1}$) [PYQ 2019] [Confidence: High]

Q40. Chromium metal is electroplated using an acidic solution containing CrO_3 according to the following equation: $\text{CrO}_3(\text{aq}) + 6\text{H}^+ + 6\text{e}^- \rightarrow \text{Cr}(\text{s}) + 3\text{H}_2\text{O}$ Calculate how many grams of chromium will be electroplated by **24,000 coulombs**. (Atomic mass of $\text{Cr} = 52 \text{ g mol}^{-1}$, $1 \text{ F} = 96500 \text{ C mol}^{-1}$) [PYQ 2019] [Confidence: High]

Q41. Using the E° values, predict which is better for coating the surface of iron to prevent corrosion and why? Given: $E^\circ_{\text{Fe}^{2+}/\text{Fe}} = -0.44 \text{ V}$, $E^\circ_{\text{A}^{2+}/\text{A}} = -2.37 \text{ V}$, $E^\circ_{\text{B}^{2+}/\text{B}} = +0.14 \text{ V}$. [PYQ 2016] [Confidence: High]

Q42. A solution of $\text{Ni}(\text{NO}_3)_2$ is electrolysed between platinum electrodes using a current of **5 amperes** for **20 minutes**. What mass of Ni is deposited at the cathode? (Molar mass of $\text{Ni} = 58.7 \text{ g mol}^{-1}$, $1 \text{ F} = 96500 \text{ C mol}^{-1}$) [PYQ 2014] [Confidence: Very High]

5-Mark Questions (Long Answer / Case-Study)

Q43. (A) Define molar conductivity. Explain how molar conductivity varies with dilution for a strong electrolyte and a weak electrolyte. (B) Calculate the standard cell potential (E°_{cell}) for the given cell reaction: $2\text{Al}(\text{s}) + 3\text{Cu}^{2+}(0.01 \text{ M}) \rightarrow 2\text{Al}^{3+}(0.01 \text{ M}) + 3\text{Cu}(\text{s})$ Given that the cell potential under these conditions (E_{cell}) is **1.98 V** at 298 K. [PYQ 2016 / Practice] [Confidence: High]

Q44. Case Study: Lead Storage Battery The lead-acid battery represents the oldest rechargeable battery technology. The spongy lead acts as the anode and lead dioxide as the cathode. Aqueous sulphuric acid is used as an electrolyte. (A) Which specific material acts as the cathode in a fully charged lead storage battery? (1 Mark) (B) What happens to the concentration/density of the sulphuric acid electrolyte during the discharging phase of the battery? (1 Mark) (C) Write the separate half-cell reactions occurring at the anode and the cathode during the **discharging** of a lead storage battery. (3 Marks) [CBSE SQP 2023] [Confidence: Very High]

Q45. Case Study: Electrochemical Theory of Rusting Corrosion is an electrochemical phenomenon. The oxygen in moist air reacts as follows: $\text{O}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l}) + 4\text{e}^- \rightarrow 4\text{OH}^-(\text{aq})$. (A) Write the specific half-cell oxidation reaction that takes place at the anode during the rusting of iron. (1 Mark) (B) Write the specific half-cell reduction reaction that takes place at the cathode during the rusting of iron. (1 Mark) (C) How does the presence of an electrolyte, such as NaCl in seawater, affect the rate of rusting? Explain the concept of 'Galvanization' as a method to prevent rusting. (3 Marks) [CBSE SQP 2022] [Confidence: Very High]

Q46. (A) Define a fuel cell. Write the overall cell reaction for a Hydrogen-Oxygen fuel cell. (B) Calculate the equilibrium constant (K_c) for the following standard cell reaction at 298 K: $\text{Cd}^{2+}(\text{aq}) + \text{Zn}(\text{s}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{Cd}(\text{s})$ Given: $E^\circ_{\text{Cd}^{2+}/\text{Cd}} = -0.403 \text{ V}$, $E^\circ_{\text{Zn}^{2+}/\text{Zn}} = -0.763 \text{ V}$, $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$, $1 \text{ F} = 96500 \text{ C mol}^{-1}$. [PYQ 2019] [Confidence: High]

Q47. (A) Write the Nernst equation for the cell reaction: $\text{Al}(\text{s}) + \text{FeSO}_4(\text{aq}) \rightarrow \text{Al}_2(\text{SO}_4)_3(\text{aq}) + \text{Fe}(\text{s})$ (B) A galvanic cell has $E_{\text{cell}} = 2.71 \text{ V}$. What will happen if an external opposing potential of (i) **2.0 V** and (ii) **3.0 V** is applied to this cell? Write the direction of electron flow in each case. (C) Why is alternating current (AC) used instead of direct current (DC) in measuring the resistance of an electrolytic solution? [PYQ 2019 / 2020] [Confidence: Very High]

Q48. Case Study: Conductivity of Aqueous Solutions Conductivity meter is a device used to test the conductivity of solutions. It has two LEDs (one red, one green). The brightness of the LEDs indicates the strength of the electrolyte. Scale: 0 (Off, Off) - None; 1 (Dim, Off) - Low; 2 (Medium, Off) - Medium; 3 (Bright, Dim) - High; 4 (Very Bright, Medium) - Very High. (A) Is it possible to identify whether a given solution is **1 M NaOH** or **1 M HCl** using this conductivity meter? Justify your answer. (1 Mark) (B) Predict whether the electrolyte is a strong or weak electrolyte on the basis of the following observation: Solution A = Scale 3, Solution B = Scale 2. (1 Mark) (C) Draw the theoretical molar conductivity (Λ_m) vs. $\sqrt{c}$ curve for Solution A and Solution B. (3 Marks) [CBSE SQP 2025-26] [Confidence: High]

Q49. (A) What is a secondary battery? Give one example. (B) The electrical resistance of a column of **0.05 mol L⁻¹ NaOH** solution of diameter **1 cm** and length **50 cm** is **5.55 × 10³ Ω**. Calculate its resistivity (ρ), conductivity (κ), and molar conductivity (Λ_m). [Practice] [Confidence: High]

Q50. (A) State Faraday's Second Law of Electrolysis. (B) Three electrolytic cells A, B, C containing solutions of ZnSO_4 , AgNO_3 and CuSO_4 , respectively, are connected in series. A steady current of **1.5 amperes** was passed through them until **1.45 g** of silver deposited at the cathode of cell B. How long did the current flow? What mass of copper and zinc were deposited? (Atomic masses: $\text{Ag} = 108$, $\text{Cu} = 63.5$, $\text{Zn} = 65.3$) [Practice] [Confidence: Very High]

Answer Key

Q1. (b) **0.1 M HCl**. (Highest concentration of strong electrolyte yields the highest total number of ions per unit volume).
Q2. (a) **0.01 M**. (Molar conductivity increases with a decrease in concentration/dilution). **Q3.** (b) **2 F**. (Reaction: $Zn^{2+} + 2e^- \rightarrow Zn$). **Q4.** (c) **5 F**. (*Mn* goes from +7 to +2, requiring 5 moles of electrons). **Q5.** (b) **193000 C**. (Reaction: $H_2O \rightarrow \frac{1}{2}O_2 + 2H^+ + 2e^-$. 2 moles of $e^- = 2 F = 2 \times 96500 = 193000 C$). **Q6.** (d) Both (a) and (c) can be used.
 $\Lambda_m^\circ CH_3COOH = \Lambda_m^\circ HCl + \Lambda_m^\circ CH_3COOK - \Lambda_m^\circ KCl$. **Q7.** (a) $\kappa = G^*/R$. **Q8.** (b) $S cm^2 mol^{-1}$. **Q9.** (a) Cathode to Anode. (Since $E_{ext} > E_{cell}$, the cell acts as an electrolytic cell and current flows in the reverse direction). **Q10.** The overall cell reaction does not involve any ion in solution whose concentration can change during its life time. **Q11.** Due to the overpotential (overvoltage) required for the oxidation of water to oxygen gas, the oxidation of chloride ions to chlorine gas is kinetically preferred. **Q12.** Hydrogen-Oxygen Fuel Cell. **Q13.** $Zn(s)|Zn^{2+}(aq)||Cu^{2+}(aq)|Cu(s)$. **Q14.** $1 F \approx 96500 C mol^{-1}$. It represents the total electrical charge carried by one mole of electrons. **Q15.** cm^{-1} or m^{-1} .
Q16. (a) Both (A) and (R) are true and (R) is the correct explanation of (A). **Q17.** (a) Both (A) and (R) are true and (R) is the correct explanation of (A). **Q18.** (d) (A) is false but (R) is true. (At equilibrium, $E_{cell} = 0$, but E_{cell}° is constant and non-zero). **Q19.** No. Both are strong electrolytes of the same concentration, so both will cause the LEDs to glow very brightly (Scale 4). They cannot be distinguished. **Q20.** **0.1 M HCl**. It is a strong electrolyte and dissociates completely into ions, whereas CH_3COOH is a weak electrolyte and dissociates partially.
Q21. The mass of any substance deposited or liberated at any electrode is directly proportional to the quantity of electricity passed. $W = Z \cdot I \cdot t$. **Q22.** Electrochemical cell: Converts chemical energy to electrical energy; $\Delta G < 0$ (spontaneous). Electrolytic cell: Converts electrical energy to chemical energy; $\Delta G > 0$ (non-spontaneous). **Q23.** Conductivity (κ) decreases because the number of ions per unit volume decreases. Molar conductivity (Λ_m) increases because the total volume containing one mole of electrolyte increases much faster than the decrease in κ . **Q24.** Degree of dissociation (α) is the fraction of total electrolyte molecules that dissociate into ions. $\alpha = \Lambda_m/\Lambda_m^\circ$. **Q25.** Mercury Cell. Anode: $Zn(Hg) + 2OH^- \rightarrow ZnO(s) + H_2O + 2e^-$. Cathode: $HgO + H_2O + 2e^- \rightarrow Hg(l) + 2OH^-$. **Q26.** Galvanic cells that convert the energy of combustion of fuels directly into electrical energy. Advantages: High efficiency (70-75%) and pollution-free. **Q27.** $E_{cell}^\circ = 0.34 - (-0.76) = 1.10 V$.
 $\Delta G^\circ = -nFE^\circ = -2 \times 96500 \times 1.10 = -212.3 kJ mol^{-1}$. **Q28.** The limiting molar conductivity of an electrolyte can be represented as the sum of the individual contributions of the anion and cation. It allows calculating Λ_m° for weak electrolytes using data from strong electrolytes. **Q29.**
 $\Lambda_m^\circ(MgCl_2) = \lambda_{Mg^{2+}}^\circ + 2 \times \lambda_{Cl^-}^\circ = 106.0 + 2(76.3) = 258.6 S cm^2 mol^{-1}$. **Q30.** $\Delta G = -nFE$. $Al^{3+} \rightarrow Al$ is $3e^-$ per Al . For $4/3 Al$, $n = 4$. $960,000 = -4 \times 96500 \times E \rightarrow E = -2.48 V$. Min potential difference = **2.48 V**. **Q31.** A is a strong electrolyte, B is a weak electrolyte. For A, Λ_m can be extrapolated to zero concentration to find Λ_m° . For B, the curve becomes parallel to the y-axis, so Λ_m° cannot be determined by extrapolation. **Q32.** Yes, a concentration cell can be constructed if the two half-cells have $ZnSO_4$ solutions of different concentrations.
Q33. $E_{cell}^\circ = 0.80 - (-0.76) = 1.56 V$.
 $E_{cell} = 1.56 - \frac{0.0591}{2} \log \frac{[0.1]}{[0.01]^2} = 1.56 - \frac{0.0591}{2} \log(1000) = 1.56 - 0.08865 = 1.47 V$. **Q34.**
 $E_{cell}^\circ = 0.80 - (-0.25) = 1.05 V$. Max work (ΔG°) = $-2 \times 96500 \times 1.05 = -202650 J = -202.65 kJ$.
 $\log K_c = \frac{nE^\circ}{0.0591} = \frac{2 \times 1.05}{0.0591} = 35.53$. **Q35.** Cell constant $G^* = \kappa \times R = 1.29 \times 10^{-2} \times 100 = 1.29 cm^{-1}$. Conductivity of $0.02M$ (κ_2) = $G^*/R_2 = 1.29/520 = 2.48 \times 10^{-3} S cm^{-1}$.
 $\Lambda_m = \kappa \times 1000/C = 2.48 \times 10^{-3} \times 1000/0.02 = 124 S cm^2 mol^{-1}$. **Q36.**
 $\Lambda_m^\circ(CH_3COOH) = \Lambda_m^\circ(HCl) + \Lambda_m^\circ(CH_3COONa) - \Lambda_m^\circ(NaCl) = 425.9 + 91.0 - 126.4 = 390.5 S cm^2 mol^{-1}$.
 $\Lambda_m = \frac{8.0 \times 10^{-5} \times 1000}{0.0024} = 33.33 S cm^2 mol^{-1}$. $\Lambda_m^\circ = 349.6 + 40.9 = 390.5 S cm^2 mol^{-1}$.
 $\alpha = 33.33/390.5 = 0.0853$ (or 8.53%). **Q38.** $Q = I \times t = 2 \times 15 \times 60 = 1800 C$. $Ag^+ + 1e^- \rightarrow Ag$.
 $Mass = \frac{108}{96500} \times 1800 = 2.014 g$. **Q39.** $Fe^{2+} + 2e^- \rightarrow Fe$. $m = \frac{M}{nF} It \Rightarrow 2.8 = \frac{56}{2 \times 96500} \times 2 \times t \Rightarrow t = 4825 s$. For Zn: $m_{Zn} = \frac{65.3}{2 \times 96500} \times 2 \times 4825 = 3.265 g$. **Q40.** Cr is $6e^-$. $Mass = \frac{52}{6 \times 96500} \times 24000 = 2.155 g$. **Q41.** Metal A (-2.37 V) is better. Its reduction potential is more negative than Fe (-0.44 V), so A acts as an anode and sacrifices itself to protect Fe (Sacrificial protection/galvanization). **Q42.** $Ni^{2+} + 2e^- \rightarrow Ni$. $Q = 5 \times 20 \times 60 = 6000 C$. $Mass = \frac{58.7}{2 \times 96500} \times 6000 = 1.82 g$.

Q43. (A) Molar conductivity (Λ_m) is the conductance of a volume of solution containing 1 mole of electrolyte placed between two electrodes unit distance apart. For strong electrolytes, it increases slowly with dilution due to reduced interionic attractions. For weak, it increases steeply due to increased degree of dissociation. (B)

$$E_{cell} = E_{cell}^\circ - \frac{0.0591}{6} \log \frac{[Al^{3+}]^2}{[Cu^{2+}]^3}. 1.98 = E_{cell}^\circ - \frac{0.0591}{6} \log \frac{(0.01)^2}{(0.01)^3}.$$

$1.98 = E_{\text{cell}}^{\circ} - 0.00985 \log(100) \Rightarrow E_{\text{cell}}^{\circ} = 1.98 + 0.0197 = 1.999 \text{ V}$. **Q44.** (A) Lead dioxide (PbO_2). (B) Density decreases (water is formed, sulfuric acid is consumed). (C) Anode: $\text{Pb} + \text{SO}_4^{2-} \rightarrow \text{PbSO}_4 + 2\text{e}^-$; Cathode: $\text{PbO}_2 + \text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^- \rightarrow \text{PbSO}_4 + 2\text{H}_2\text{O}$. **Q45.** (A) Anode: $\text{Fe(s)} \rightarrow \text{Fe}^{2+}(\text{aq}) + 2\text{e}^-$. (B) Cathode: $\text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4\text{e}^- \rightarrow 2\text{H}_2\text{O(l)}$. (C) Ions in the electrolyte (NaCl) increase the conductivity of the water, accelerating electron flow and thus the rate of rusting. Galvanization is coating iron with a more reactive metal (like Zn) that oxidizes preferentially, protecting the iron. **Q46.** (A) Fuel cell is a device that converts combustion energy of fuels directly into electricity. Reaction: $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O(l)}$. (B) $E_{\text{cell}}^{\circ} = -0.403 - (-0.763) = 0.36 \text{ V}$. $\log K_c = \frac{nE^{\circ}}{0.0591} = \frac{2 \times 0.36}{0.0591} = 12.18$. $K_c = 10^{12.18}$ or 1.51×10^{12} . **Q47.** (A) $E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{2.303RT}{6F} \log \frac{[\text{Al}^{3+}]^2}{[\text{Fe}^{2+}]^3}$. (B) (i) Current flows from anode to cathode (normal galvanic). (ii) Current flows from cathode to anode (acts as an electrolytic cell). (C) DC causes electrolysis/polarization, changing the concentration of the solution near the electrodes. AC prevents this. **Q48.** (A) No, both are strong electrolytes and fully dissociate, producing maximum brightness (Scale 4). (B) A is a strong electrolyte, B is a weak electrolyte. (C) Graph: A is a straight line with a slight negative slope. B is a curve that rises steeply near $\sqrt{c} = 0$. **Q49.** (A) A cell that can be recharged by passing current in the opposite direction. Example: Lead storage battery. (B) Area $A = \pi r^2 = 3.14 \times (0.5)^2 = 0.785 \text{ cm}^2$. $R = \rho \frac{l}{A} \Rightarrow 5550 = \rho \frac{50}{0.785} \Rightarrow \rho = 87.135 \Omega \text{ cm}$. $\kappa = \frac{1}{\rho} = 0.01148 \text{ S cm}^{-1}$. $\Lambda_m = \frac{\kappa \times 1000}{C} = \frac{0.01148 \times 1000}{0.05} = 229.6 \text{ S cm}^2 \text{ mol}^{-1}$. **Q50.** (A) The masses of different substances deposited by the same quantity of electricity are proportional to their equivalent weights. (B) Mass of $\text{Ag} = \frac{108}{1 \times 96500} \times 1.5 \times t \Rightarrow 1.45 = \frac{108 \times 1.5 \times t}{96500} \Rightarrow t = 863.7 \text{ s}$. Mass of $\text{Cu} (2\text{e}^-) = \frac{63.5}{2 \times 96500} \times 1.5 \times 863.7 = 0.426 \text{ g}$. Mass of $\text{Zn} (2\text{e}^-) = \frac{65.3}{2 \times 96500} \times 1.5 \times 863.7 = 0.438 \text{ g}$.

CBSE Class 12 (2027) — Chapterwise PYQ Bank. Compiled from past board papers + board-pattern practice questions (clearly labelled).